

Quiz 3-B

First and Last Name: _____

You have 15 minutes to complete the quiz. Calculators, notes, and other resources are **not** allowed.

1. (4 points) Find A if $(I_2 + 3A)^{-1} = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$. Show all work. Your final answer should be an explicit 2×2 matrix, but you don't need to simplify the entries.

Solution: Apply the formula for inverse of 2×2 matrices, we get

$$\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}^{-1} = \frac{1}{3 \cdot 2 - 2 \cdot 1} \begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/4 & 3/4 \end{bmatrix}.$$

Then $I_2 + 3A = \begin{bmatrix} 1/2 & -1/2 \\ -1/4 & 3/4 \end{bmatrix}$, we solve

$$A = \frac{1}{3} \left(\begin{bmatrix} 1/2 & -1/2 \\ -1/4 & 3/4 \end{bmatrix} - I_2 \right) = \begin{bmatrix} -1/6 & -1/6 \\ -1/12 & -1/12 \end{bmatrix}.$$

QUESTION 2 IS ON THE BACK!!

2. Suppose A is a 3×3 matrix. Answer each question. No work/justification is necessary.

Note: r_i represents row i .

- (a) (1 point) Find the 3×3 matrix E where EA is the same as performing the following elementary row operation to A : r_1 is swapped with r_2 .

Solution: Apply the same operation to I_3 : r_1 is swapped with r_2 , we get

$$E = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- (b) (1 point) Find the 3×3 matrix F where FA is the same as performing the following elementary row operation to A : Replace r_3 with $5r_1 + r_3$.

Solution: Apply the same operation to I_3 : Replace r_3 with $5r_1 + r_3$, we get

$$F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 5 & 0 & 1 \end{bmatrix}.$$